

No. of Functional Features Included in the Solution

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Flood Risk Prediction	Predicts flood likelihood based on 20 environmental input parameters	app.py, flood_model.pkl	Done	1 Mark
2	ML Model Training	Trains four classifiers and selects the best performing model automatically	train_model.py	Done	1 Mark
3	Input Data Form	Web form accepting 20 meteorological and environmental parameter values from user	index.html	Done	0.5 Mark
4	Prediction Result Page	Displays flood risk result with color coded banner and risk level indicator bar	result.html	Done	0.5 Mark
5	Recommendations System	Shows actionable steps based on whether flood is likely or not likely	result.html	Done	0.5 Mark
6	Model Persistence	Saves trained XGBoost model as pkl file and loads it for every prediction	joblib, model/	Done	0.5 Mark
7	About Page	Displays project information, technology stack and model accuracy comparison table	about.html	Done	0.5 Mark
8	Responsive Navigation	Navigation bar with links to Home and About pages across all pages	index, result, about.html	Done	0.5 Mark

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	4	Input form, result page, about page, navigation bar
2	Backend / Logic	3	Flood prediction, ML model training, model persistence
3	Database / Storage	1	Saving and loading trained model as flood_model.pkl
4	API / Integration	0	Not implemented in current version
5	Security / Authentication	0	Not implemented in current version